

Prof. Dr. Johannes Freudenmann (DHBW Karlsruhe)
Masterprogramm Informatik

DNS-Scanner

www.dhbw.de

DNS-Scanner

Anwender nutzen viele Dienste im Internet.
Probleme mit Malware nehmen immer mehr zu.
Wie kann ich meine Anwender besser schützen?

- URL – Filter.

Probleme:

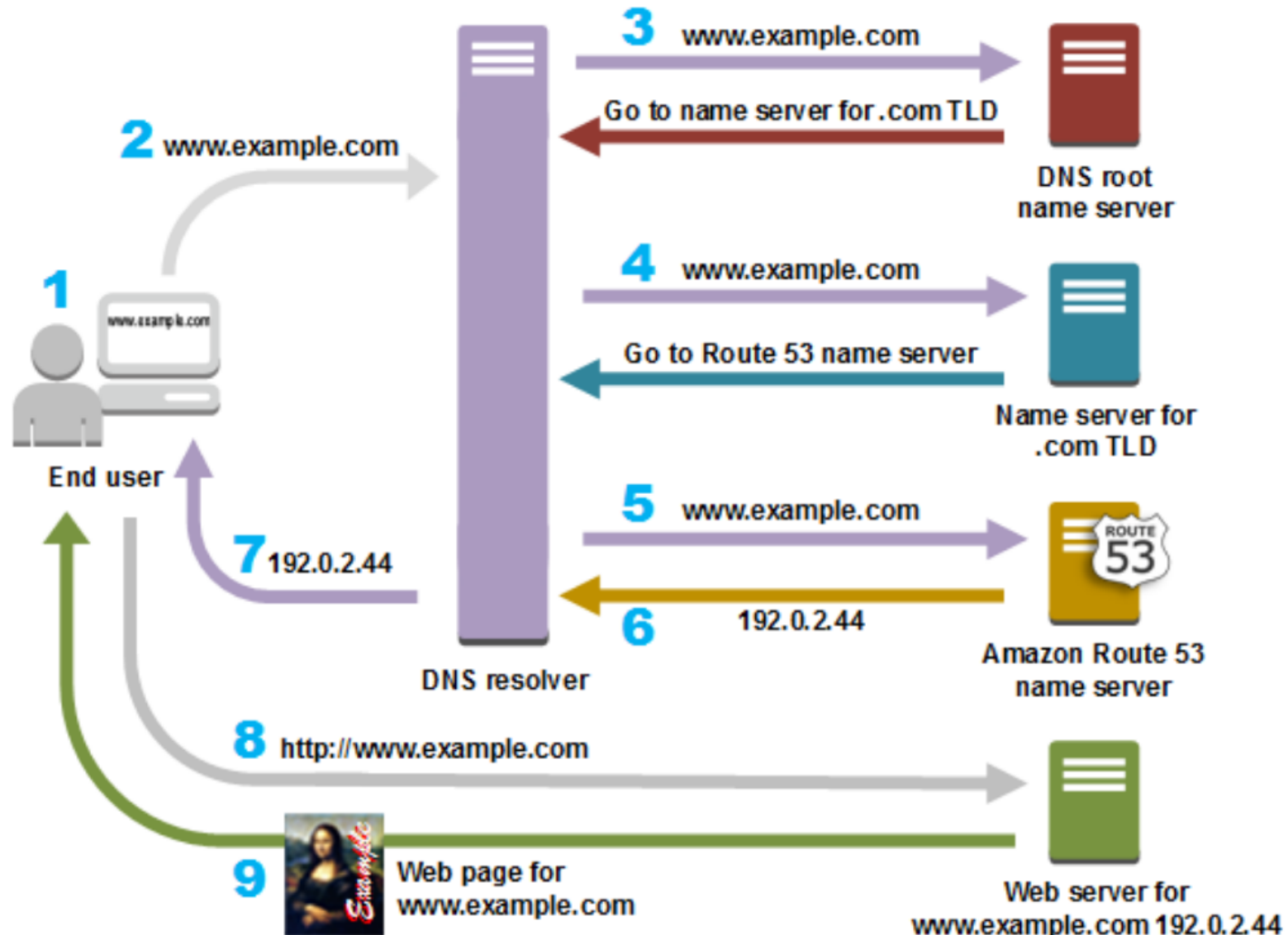
- Schutz nur für Webserver
- Probleme bei Verschlüsselung

Alternative: Kopplung an DNS.

- Alternativer DNS-Server wird konfiguriert
- Bsp. Umbrella von Cisco
- Dienst auch von anderen Stellen verfügbar.

DNSSense, DFN/DNS-RPZ, ????

Erläuterung DNS (Erinnerung)



Umbrella Security Settings

- ☒ Malware
Websites and other servers that host malicious software, drive-by downloads/exploits, mobile threats and more.
- ☒ Newly Seen Domains
Domains that have become active very recently. These are often used in new attacks.
- ☒ Command and Control Callbacks
Prevent compromised devices from communicating with attackers' infrastructure.
- ☒ Phishing Attacks
Fraudulent websites that aim to trick users into handing over personal or financial information.
- ☐ Dynamic DNS
Block sites that are hosting dynamic DNS content.
- ☐ Potentially Harmful Domains
Domains that exhibit suspicious behavior and may be part of an attack.
- ☐ DNS Tunneling VPN
VPN services that allow users to disguise their traffic by tunneling it through the DNS protocol. These can be used to bypass regarding access and data transfer.
- ☒ Cryptomining
Cryptomining allows organizations to control cryptominer access to mining pools and web miners.

Umbrella Content selection

Categories [SELECT ALL](#)

- | | | |
|---|---|--|
| ☐ Adult | ☐ Hate Speech | ☐ Politics |
| ☐ Advertisements | ☐ Health and Medicine | ☐ Pornography |
| ☐ Alcohol | ☐ Humor | ☐ Private IP Addresses as Host |
| ☐ Animals and Pets | ☐ Hunting | ☐ Professional Networking |
| ☐ Arts | ☐ Illegal Activities | ☐ Real Estate |
| ☐ Astrology | ☐ Illegal Downloads | ☐ Recipes and Food |
| ☐ Auctions | ☐ Illegal Drugs | ☐ Reference |
| ☐ Business and Industry | ☐ Infrastructure and Content Delivery Networks | ☐ Regional Restricted Sites (Germany) |
| ☐ Cannabis | ☐ Internet of Things | ☐ Regional Restricted Sites (Great Britain) |
| ☐ Chat and Instant Messaging | ☐ Internet Telephony | ☐ Regional Restricted Sites (Italy) |
| ☐ Cheating and Plagiarism | ☐ Job Search | ☐ Regional Restricted Sites (Poland) |
| ☐ Cloud and Data Centers | ☐ Lingerie and Swimsuits | ☐ Religion |
| ☐ Computer Security | ☐ Lotteries | ☐ SaaS and B2B |
| ☐ Computers and Internet | ☐ Missing | ☐ Safe for Kids |

Umbrella Meldung





Diese Seite ist aufgrund einer Phishing-Bedrohung blockiert.

www.internetbadguys.com

Phishing ist der betrügerische Versuch, unter Vorspiegelung falscher Tatsachen persönliche Informationen von Ihnen zu erhalten.

PLEASE READ CAREFULLYAccess to this web page is blocked in accordance with ATS Security Policies. If you believe this page you are attempting to access is appropriate and is improperly rated, please contact your IT Administrator.

Diese Website wurde aufgrund der folgenden Kategorien blockiert: **Phishing**

[> Diagnoseinformationen](#)

Umbrella Anmerkungen

- Realisierbar einfach durch Eintrag des DNS-Server.
- Beschränkt auf die Auswahl von Domänen. Keine Differenzierung zwischen verschiedenen Webseiten auf demselben Server.
- Positivlisten, Negativlisten, Künstliche Intelligenz, ... einsetzbar.
- Policy abhängig vom Betreiber.
- Ergänzung
 - Lokaler Client -> differenzierte weitere Auswahl möglich